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Editorial Board
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Dear Editors,

We submit the accompanying manuscript, “Algorithm suitability for photonic continuous-variable quantum computers: a comparative resource and noise-robustness study,” for consideration as a Letter in *Applied Physics Express*. The work falls within APEX’s scope of applied photonics and quantum technologies, and it addresses a question of direct practical relevance to the design of GKP-based photonic processors that several groups are now constructing experimentally.

Motivation. Photonic continuous-variable (CV) architectures support deterministic, room-temperature generation of large cluster states and are a leading route to fault-tolerant quantum computing. Their cost, however, is dominated by squeezing and by the consumption of non-Gaussian resource states. Different quantum algorithms place very different demands on these resources, and no unified comparison has quantified which algorithm classes are intrinsically robust on CV hardware and how much GKP error correction actually helps. The present work supplies that comparison and the design rules that follow from it.

Main findings. We port four representative algorithms—control-free phase estimation, randomized product formulas, quantum Kolmogorov–Arnold networks, and Regev factoring—to a common CV model and compare them under photon loss (exact Fock-space Kraus channel) and Gaussian displacement with GKP correction. Three results are central: (i) control-free phase estimation is the most noise-robust because it needs no controlled unitaries and, for quadratic Hamiltonians, no non-Gaussian gates, and the property survives for non-Gaussian Hamiltonians; (ii) GKP correction reduces residual error only above the finite-squeezing floor, and whether an algorithm benefits is set by a noise- versus systematic-limited dichotomy whose empirical envelope constants we report with $R^2 \geq 0.89$; (iii) against a qubit surface-code baseline, the CV advantage is structural—Gaussian operations are free—rather than threshold-based.

Quantitative novelty. The core GKP-correction model is not assumed: it is validated against an explicit Glancy–Knill error-correction Monte Carlo to within 0.1%, lifting a common methodological concern in CV-resource papers. A two-dimensional phase diagram and Hamiltonian ensembles confirm the predictions are not artefacts of single operating points or single instances. We are explicit, in a dedicated paragraph, about which results are exact, which are validated model surrogates, and which are extrapolations.

Experimental relevance. The natural test bed is a GKP-encoded photonic cluster-state processor at 10–15 dB squeezing—within reach of current laboratories—where the predicted floor crossing and the algorithm dichotomy are directly observable; control-free spectroscopy of a small bosonic Hamiltonian is the most accessible first demonstration. We believe this makes the manuscript a

timely and useful contribution to APEX's photonics readership.

Compliance. The Letter is below the APEX length and figure limits, the abstract is within 100 words, and the manuscript carefully avoids the discouraged language patterns. All scripts and numerical results that reproduce every figure and table are bundled with the submission. The manuscript has not been published and is not under consideration elsewhere; the authors declare no competing interests.

Sincerely,

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